FIITJEE

ALL INDIA TEST SERIES

FULL TEST – XVIII

JEE (Main)-2021

TEST DATE: 21-08-2021

Time Allotted: 3 Hours **Maximum Marks: 300**

General Instructions:

- The test consists of total 90 questions.
- Each subject (PCM) has 30 questions.
- This question paper contains **Three Parts.**
- **Part-A** is Physics, **Part-B** is Chemistry and **Part-C** is Mathematics.
- Each part has only two sections: **Section-A and Section-B**.
- **Section – A** : Attempt all questions.
- **Section – B :** Do any five questions out of 10 Questions.

  Section-A (01 – 20, 31 – 50, 61 – 80) *contains 60 multiple choice questions which have* ***only one correct answer****. Each question carries* ***+4 marks*** *for correct answer and* ***–1 mark*** *for wrong answer.*

  Section-B (21 – 30, 51 – 60, 81 – 90) *contains 30 Numerical answer type questions with answer XXXXX.XX and each question carries* ***+4 marks*** *for correct answer. There is no negative marking.*

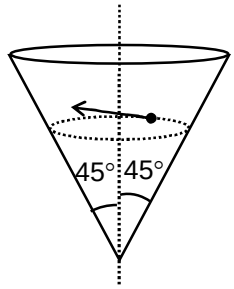
Physics

PART – A

SECTION – A
(One Options Correct Type)

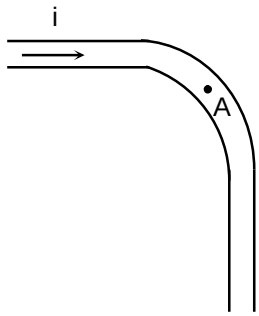
This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

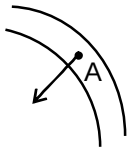
1. A particle is moving in horizontal circle inside a smooth vertical cone under the influence of gravity and normal. Now the particle is hit by some external agent so that its speed becomes double in the same direction. The acceleration of particle just after the hit is:

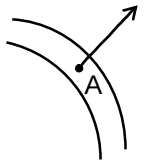


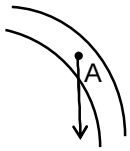
(A) g
(B) 3g
(C) 4g
(D) 5g

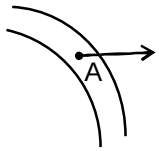
2. A steady current flows in the conductor of uniform cross sectional area in the direction shown. Then the direction of electric field at point A inside the conductor is



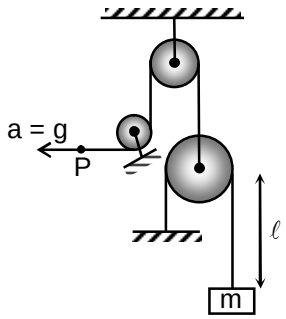
(A)


(B)


(C)


(D)


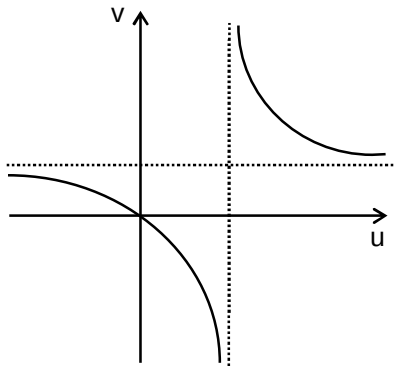
3. The free end P of the string is pulled towards left with a constant acceleration g from rest. Then the work to be done in pulling the point P until the block reaches the lower pulley is (All the pulleys and strings are ideal and massless).



(A) $4mg\ell$
(B) $5mg\ell$
(C) $6\ mg\ell$
(D) $7\ mg\ell$

4. The u-v graph shown in the diagram is correctly match for

(A) Converging mirror
(B) Diverging mirror
(C) Converging lens
(D) Diverging lens



5. The ratio of efficiency of Carnot's engine to the coefficient of performance of Carnot's refrigerator, if both are operating in between T_1 and T_2 is ($T_1 > T_2$)

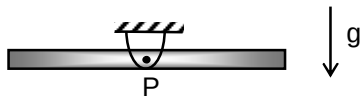
(A) $\frac{(T_1 - T_2)^2}{T_1 T_2}$

(B) $\frac{(T_1 - T_2)^2}{T_1^2}$

(C) $\frac{T_1 T_2}{(T_1 - T_2)^2}$

(D) $\frac{T_2^2}{(T_1 - T_2)^2}$

6. A uniform rod of length L is hinged from a point P and released in the position when it is horizontal. Find the maximum angular acceleration of the rod at this instant.



(A) $\frac{3g}{L}$

(B) $\frac{3g}{2L}$

(C) $\frac{\sqrt{3}g}{L}$

(D) $\frac{\sqrt{3}g}{2L}$

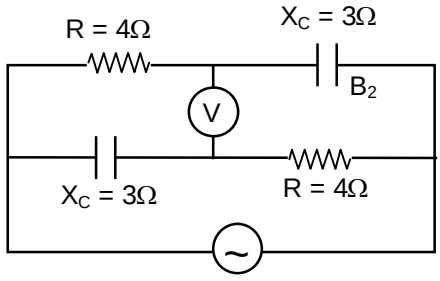
7. The phase difference between the source voltage and the voltage measured by the A.C. voltmeter is

(A) zero

(B) $\frac{\pi}{2}$

(C) $2\tan^{-1}\left(\frac{3}{4}\right)$

(D) $2\tan^{-1}\left(\frac{4}{3}\right)$



8. The images of an object for two different object distances u_1 and u_2 are I_1 and I_2 respectively. If size of I_2 is twice of size of I_1. Then the focal length of the lens satisfies.

(A) $3f - u_1 - 2u_2 = 0$

(B) $3f - u_1 + 4u_2 = 0$

(C) $3f + u_1 - 2u_2 = 0$

(D) $3f + u_1 + 2u_2 = 0$

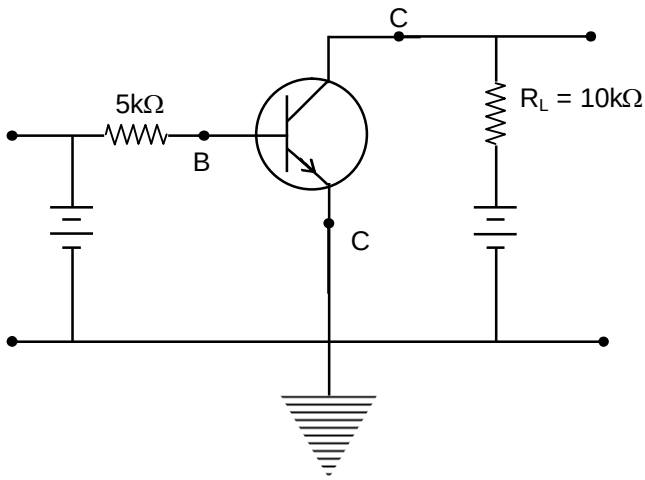
9. The diagram shows common emitter configuration used in an amplifier. 98% of the charge carriers passing through emitter reach the collector. If base current changes from 50 μA to 100 μA, then the corresponding change in the voltage across the load resistance R_L will be

(A) 0.25 V

(B) 0.5 V

(C) 24.5 V

(D) 49.0 V



10. A charged particle of mass m and charge e moves in a circular path of radius r under the action of a uniform magnetic field perpendicular to the plane of the circular path. If Bohr's postulate is valid for this system, then radius of the path is given by

(A) $r \propto n^2$

(B) $r \propto \sqrt{n}$

(C) $r \propto \frac{1}{n}$

(D) $r \propto \frac{1}{\sqrt{n}}$

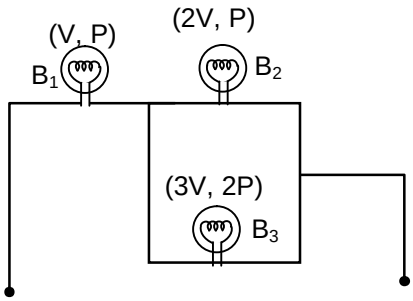
11. The combination of three bulbs as shown whose rated voltage and rated power given in brackets is equivalent to a single bulb whose rated voltage and rated power are given as V, P_{eq}. Then P_{eq} is

(A) $\frac{17P}{53}$

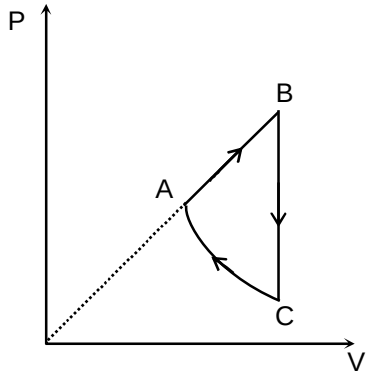
(B) $\frac{3P}{4}$

(C) $\frac{53P}{17}$

(D) $\frac{4P}{3}$



12. The ideal monoatomic gas undergoes a cyclic process ABCA in which BC and CA are isochoric and isothermal. If ratio of maximum to minimum volume during the process is 2. Then efficiency of the cyclic process is



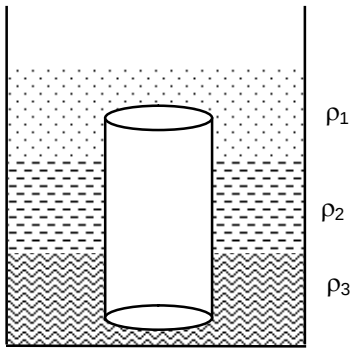
(A) $\frac{3-2\ell n2}{12}$

(B) $\frac{9-2\ell n2}{12}$

(C) $\frac{3-\ell n2}{12}$

(D) $\frac{9-\ell n2}{12}$

13. The system shown in the figure is in equilibrium. Then which of the following statements is INCORRECT.



(A) Net force applied by liquid-2 on the cylinder is zero.
(B) Net force applied by liquid-2 on the cylinder is in upward direction
(C) Net force applied by liquid-3 on the cylinder is in upward direction
(D) Net force applied by liquid-1 on the cylinder is in downward direction

14. A uniform conducting rod of length L is rotating about an axis passing through one of its end points and perpendicular to the rod with constant angular velocity ω. If charge to mass ratio of electron is γ, then potential difference across the end points of the rod is (The space is gravity free and there is no electric and magnetic field in the space)

(A) zero

(B) $\frac{\omega^2 L^2}{\gamma}$

(C) $\frac{2\omega^2 L^2}{3\gamma}$

(D) $\frac{\omega^2 L^2}{2\gamma}$

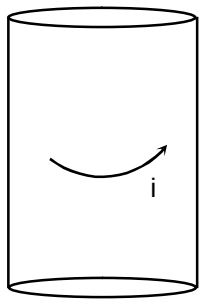
15. A solid uniform sphere of mass 1 kg having specific heat capacity of 0.1 J-kg^{-1}K^{-1} and radius 20 m having thermal coefficient of linear expansion 10^{-3} °C^{-1} is placed on a fixed horizontal surface. Q heat is supplied to the sphere and its temperature increases by 10°C. The Q is

(A) 3J
(B) 2.96 J
(C) 1 J
(D) 1.98 J

16. A parallel beam of monochromatic light is incident on a slit of width 2 cm. The light passing through the slit falls on a screen 2m away. As the slit width is decreased, then
(A) The width of the pattern on screen continuously decreases
(B) The width of the pattern on the screen first decreases and then increases.
(C) The width of the pattern on screen continuously first increases and then decreases.
(D) The width of the pattern on screen remains same.

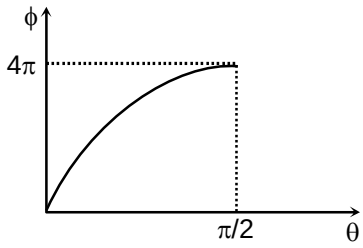
17. If the unknown resistance calculated without using the end correction (in meter bridge) is R_1 and with using the end correction is R_2, then
(A) $R_1 > R_2$, when balanced point is in first half
(B) $R_1 > R_2$, when balanced point is in second half
(C) $R_1 > R_2$, always
(D) $R_1 < R_2$, always

18. A hollow cylinder having infinite length and carrying uniform current per unit length λ along the circumference as shown. Magnetic field inside the cylinder is



(A) $\frac{\mu_0\lambda}{2}$
(B) $2\mu_0\lambda$
(C) $\mu_0\lambda$
(D) none of these

19. In YDSE setup, the plot of the phase difference (ϕ) between both waves at point P on the screen against the angular position (θ) of point P on the screen is shown below. Choose the INCORRECT statement.



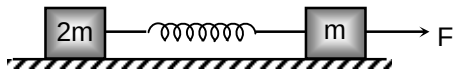
(A) The separation between slits is 2λ
(B) There are 4 minimas on the screen
(C) The first maxima above the centre is at $\theta = \frac{\pi}{4}$
(D) There are 3 maximas on the screen

20. Binding energy per nucleon of a fixed nucleus X^A is 6 MeV. It absorbs a neutron moving with K.E. = 2MeV and converts into Y at ground state, emitting a photon of energy 1 MeV. The binding energy per nucleon of Y (in MeV) is
(A) 7
(B) $\frac{6A+1}{A+1}$
(C) $\frac{7}{6}$
(D) $\frac{6A-1}{A+1}$

SECTION – B
(Numerical Answer Type)

This section contains **10** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

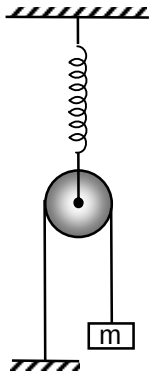
21. Two blocks of masses m and 2m are connected by an unstretched spring and the system is kept on a frictionless horizontal surface. A constant force F is applied on the block of mass m as shown in the figure. If the extension in the spring is X_0 at time t, then the displacement of the left block at this instant is $\frac{Ft^2}{6m} - \frac{X_0}{\lambda}$. Then λ is



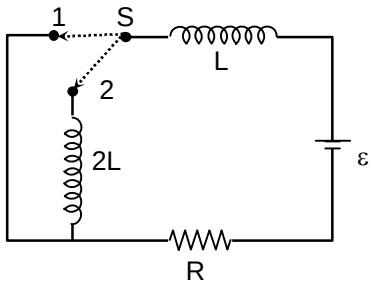
22. A continuous flow water heater (geyser) has water inlet temperature = 10°C while the water flows out at the rate of 20 g/sec. The power of geyser is 2100 watt. The outlet temperature of water is (5x)°C. Then x is

23. The minimum number of capacitors required each of capacitance 2 μF in order to obtain the resultant capacitance of 5 μF is

24. The system consists of a smooth pulley of mass 2m, a block of mass m and the massless spring as shown in the diagram is released from the position when spring is in natural length. The time after the release when the pulley comes to rest for the first time is $\pi\sqrt{\frac{\alpha m}{k}}$, then α is

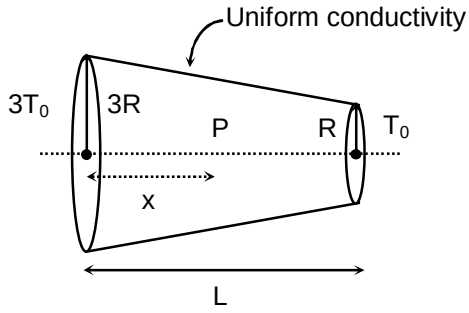


25. In the circuit shown, the switch S is shifted to position 2 from position 1 at t = 0, having been in position 1 for a long time. Find the ratio of current in the inductor (L) just before and just after the switch S is shifted.



26. The voltage applied to an x-ray tube with nickel anticathode, if the wavelength difference between the K_α and the cut-off wavelength of the continuous x-rays spectrum is equal to 84 pm, is (3n) kV. Then n is

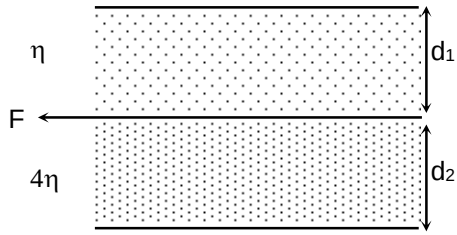
27. The ends of a frustum are maintained at temperatures $3T_0$ and T_0 as shown in the figure. Find x (in meter) if temperature at point P is $2T_0$. (length of frustum L = 12 meter)



28. Two sources A and B are producing notes of frequency 680 Hz. A detector moves from A to B with a constant velocity 2 m/s. The number of beats heard by the detector per second is (in Hz) (velocity of sound = 340 m/s)

29. A body cools from 61°C to 59°C in 4 minutes in a room temperature of 30°C. The time taken by the body to cool from 51°C to 49°C will be (in minutes) when room temperature is 30°C.

30. A thin plate separates two liquids of coefficients of viscosity η and 4η kept between two fixed plates as shown. If plate has to be pulled by applying minimum force then $\frac{d_1}{d_2}$ is



Chemistry

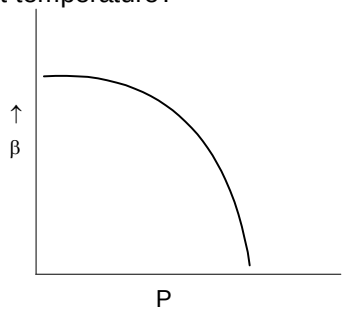
PART – B

SECTION – A
(One Options Correct Type)

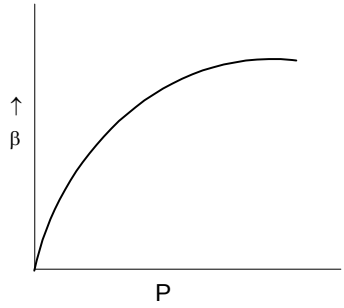
This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

31. Which of the following graphs correctly represents the variation of $\beta = -\left(\frac{dV/dP}{V}\right)$ for ideal gas at constant temperature?

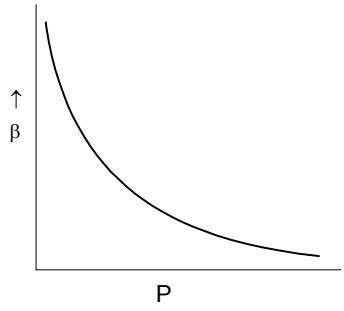
(A)



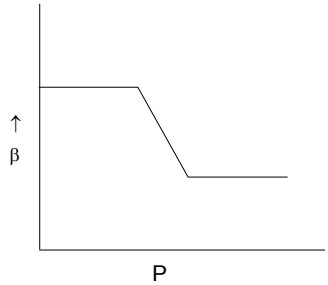
(B)



(C)



(D)



32. At 27^oC, it was observed in the hydrogenation reaction, the pressure of H_2 gas decreases from 10 atm to 2 atm in 10 mins. Calculate the rate of reaction in $M\,min^{-1}$.

(Given: $R = 0.08$ lit atm K^{-1} mol^{-1})

(A) 0.02
(B) 0.03
(C) 0.04
(D) 0.05

33. Efficiency of the following cells is 84%

$X(s) + Y^{2+}(aq) \rightleftharpoons X^{2+}(aq) + Y(s)$ $\Delta H = -285\ kJ/mol$

Then the standard electrode potential of the cell will be

(A) 1.20 V
(B) 2.40 V
(C) 1.24 V
(D) 1.10 V

34. If the equilibrium constant of the reaction of weak acid HX with strong base is 10^7 then pH of 0.1 M NaX is

(A) 9
(B) 5
(C) 10
(D) 8

35. For a particular reaction

$\Delta H^o = -38.3$ kJ and $\Delta S^o = -113\ JK^{-1}\ mol^{-1}$.

The reaction is

(A) spontaneous at all temperatures.
(B) non-spontaneous at all temperatures.
(C) spontaneous at temperature below 66^oC.
(D) spontaneous at temperature above 66^oC.

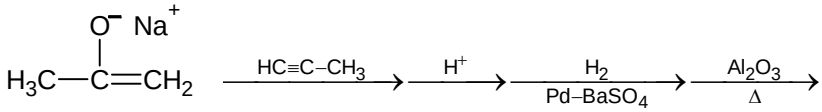
36. The vapour pressure of pure liquid solvent A is 0.80 bar. When a non-volatile substance B is added to the solvent its vapour pressure drops to 0.60 bar. What is the mole fraction of component B in the solution?

(A) 0.75
(B) 0.50
(C) 0.25
(D) 1.0

37. The de-Broglie wavelength of neutrons at 27^oC is λ. The wavelength at 927^oC will be

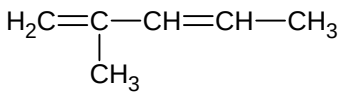
(A) $\frac{\lambda}{9}$
(B) $\frac{\lambda}{2}$
(C) $\frac{\lambda}{4}$
(D) $\frac{\lambda}{3}$

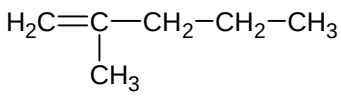
38.

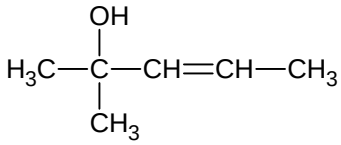


Final product of the reaction is

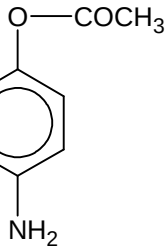
(A) $CH_3 - CH = CH - CH = CH_2$

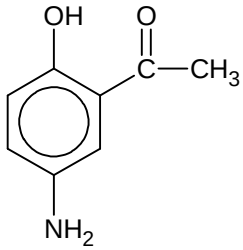
(B) 

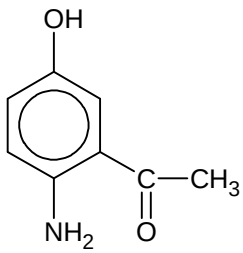
(C) 

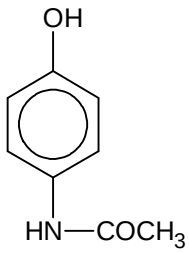
(D) 

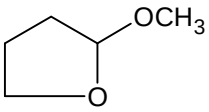
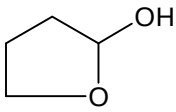
39. p-aminophenol reacts with one equivalent of acetyl chloride in presence of pyridine to give mainly:

(A) 

(B) 

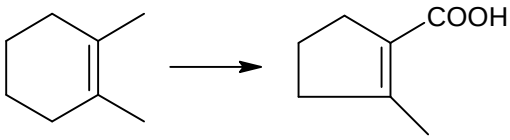
(C) 

(D) 

40.  and 

Above compounds can be differentiated by the following reagent?

(A) 2-4-DNP (Brady reagent)
(B) Tollen's reagent
(C) Lucas reagent
(D) $NaHSO_3$

41. 

In order to carry out above conversion, arrange the following reagents in correct order

O_3 / Zn (I) EtONa / EtOH / Δ (II) NaOCl (III) H^+ (IV)

(A) I – III – II – IV
(B) I – II – III – IV
(C) I – III – IV – II
(D) I – II – IV – III

42. Find out the reaction in which resultant product give +ve carbylamine test?

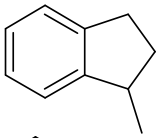
(A) $C_6H_5-\overset{\overset{O}{\|}}{C}-NH_2 \xrightarrow{NaBH_4}$

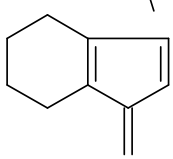
(B) $C_6H_5-NH-\overset{\overset{O}{\|}}{C}-CH_3 \xrightarrow{LiAlH_4}$

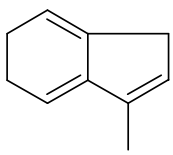
(C) $C_6H_5-\overset{\overset{O}{\|}}{C}-NH_2 \xrightarrow{LiAlH_4}$

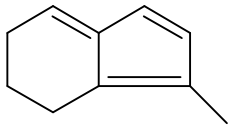
(D) $C_6H_5-NH-\overset{\overset{O}{\|}}{C}-CH_3 \xrightarrow{NaBH_4}$

43. Which of the following isomeric hydrocarbons is most acidic?

(A) 

(B) 

(C) 

(D) 

44. Reaction of R-2-butanol with p-toluene sulphonyl chloride in pyridine followed by reaction with NaBr gives
(A) R-2-butyl bromide
(B) S-2-butyl bromide
(C) S-2-butyl tosylate
(D) R-2-butyl tosylate

45. Which of the following pair of ions has same value of "spin only" magnetic moment?
(A) Cu^{+}, Cu^{2+}
(B) Co^{3+}, Fe^{2+}
(C) Ti^{2+}, Cu^{2+}
(D) Sc^{2+}, Zn^{2+}

46. The molecule exhibiting maximum number of non-bonding electron pairs (ℓp) around the central atom is
(A) $XeOF_4$
(B) XeO_2F_4
(C) XeF_3^-
(D) XeO_3

47. Holme's signal can be given by using
(A) $CaC_2 + CaCN_2$
(B) $CaC_2 + Ca_3P_2$
(C) $CaC_2 + CaO$
(D) $Ca_3P_2 + CaCN_2$

48. The increasing order of the ionic radii of the given isoelectronic species
(A) Cl^-, Ca^{2+}, K^+ and S^{2-}
(B) S^{2-}, Cl^-, Ca^{2+} and K^+
(C) Ca^{2+}, K^+, Cl^- and S^{2-}
(D) K^+, S^{2-}, Ca^{2+} and Cl^-

49. The number of P – O – P bonds in cyclic metaphosphoric acid is
(A) Zero
(B) Two
(C) Three
(D) Four

50. Among the following, the incorrect statement is
(A) Argentite and cuprite are oxide ores
(B) Calamine and azurite are carbonates
(C) Zinc blend and pyrites are sulphides
(D) Malachite and azurite are ores of coppers

SECTION – B
(Numerical Answer Type)

This section contains **10** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

51. Calculate the experimental bond moment of A – B bond if electronegativity of elements A and B is 1 and 2.5 respectively and bond length is $3.92\overset{o}{A}$.

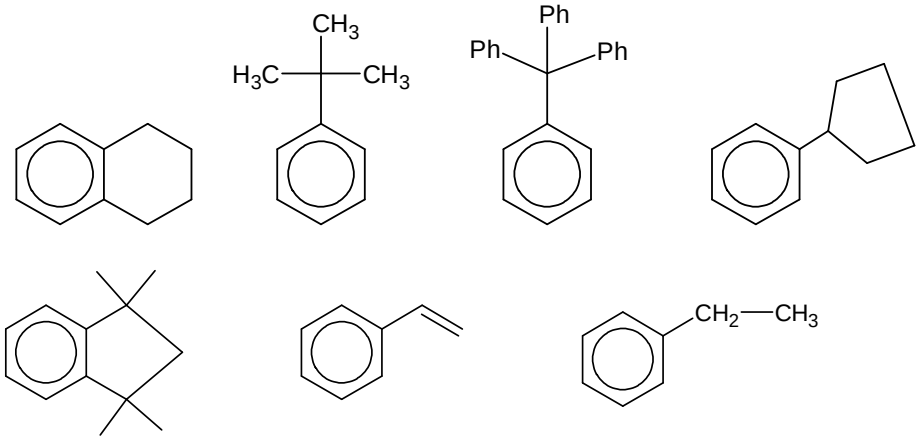
52. Find the magnetic moment of the following compound
$Na\left[Cr(NH_3)_2OX_2\right].3H_2O$.

53. 0.01 M CH_3COOH is 4.24% ionized. What will be the percentage ionization of 0.1 M CH_3COOH ?

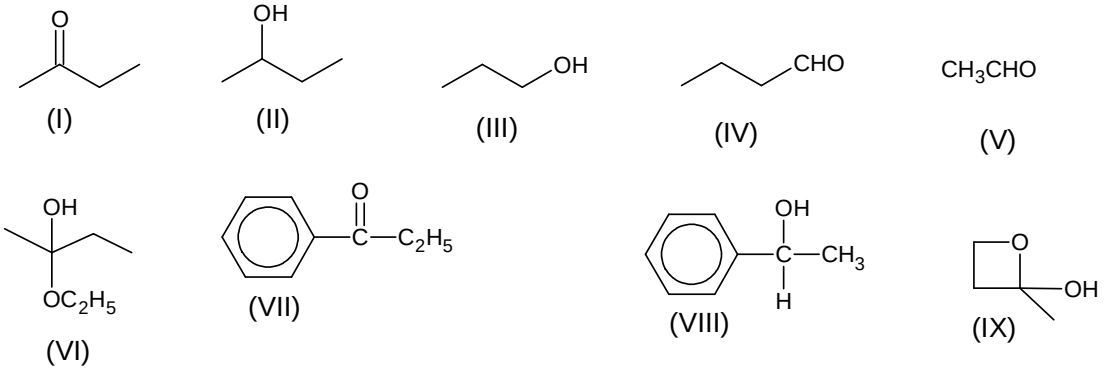
54. A 0.5 M NaOH solution offers a resistance of 31.6 Ohm in conductivity cell at room temperature. What will be the approximate molar conductance of this NaOH solution, if the cell constant of the cell is $0.367\ cm^{-1}$?

55. If the half-lives of a radioactive element for α and β decay are 4 years and 12 years respectively, then the % of element that remains after 12 years will be:

56. Examine the structural formulae shown below and find out how many compounds will show oxidation reaction with acidic $KMnO_4$.

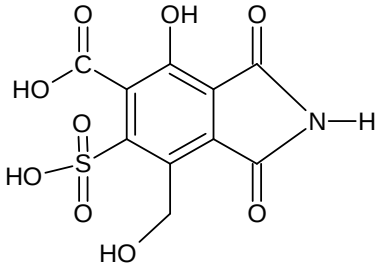


57. How many of the following compounds will give iodoform test



58. The number of tripeptides formed by three different amino acids are:

59. One mole of the following compound is treated with NaOH solution. How many moles of NaOH would be required for complete neutralization?



60. What mass of solid ammonium carbamate $\left(NH_2COONH_4\right)$, when vapourized at 273^oC will have a volume of 8.96 L at 760 mm of pressure, assuming complete decomposition as given below

$H_2NCOONH_4\left(s\right)\longrightarrow CO_2\left(g\right)+2NH_3\left(g\right)$

Mathematics

PART – C

SECTION – A
(One Options Correct Type)

This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

61. $\lim\limits_{n\to\infty}\sum\limits_{r=1}^{2n}\tan^{-1}\left(\frac{1}{2r^2}\right)$ is equal to

(A) $\frac{\pi}{2}$

(B) $\frac{\pi}{4}$

(C) 1

(D) $\frac{3\pi}{4}$

62. If the equations $ax^2 + bx + c = 0$ and $x^3 + 3x^2 + 3x + 2 = 0$ have two common roots, then

(A) $a = b = c$

(B) $a = b \neq c$

(C) $a = -b = c$

(D) $a = b = -c$

63. The greatest positive argument of complex number satisfying $|z - 4| = Re(z)$ is

(A) $\frac{\pi}{3}$

(B) $\frac{2\pi}{3}$

(C) $\frac{\pi}{2}$

(D) $\frac{\pi}{4}$

64. The value of $\sum\limits_{r=0}^{10}(-1)^r \cdot 4^{10-r} \cdot {}^{30}C_r \cdot {}^{30-r}C_{10-r}$ is equal to

(A) ${}^{30}C_{10} \times 2^{10}$

(B) ${}^{30}C_9 \times 4^{10}$

(C) ${}^{30}C_{10} \times 3^{10}$

(D) ${}^{30}C_9 \times 3^{10}$

65. The number of zeroes at the end in product $5^6 \cdot 6^7 \cdot 7^8 \cdot 8^9 \cdot 9^{10} ..... 30^{31}$ is equal to

(A) 127

(B) 137

(C) 147

(D) 157

66. If $p \rightarrow (q \vee r)$ is false, then the truth values of p, q and r respectively are
(A) F, T, T
(B) T, T, F
(C) T, F, F
(D) F, F, F

67. The standard deviation of 25 numbers is 40. If each of the numbers is increased by 5, then the new standard deviation will be
(A) 40
(B) 45
(C) 35
(D) 25

68. If $\alpha\beta\delta \neq 0$ and the system of equations $(\alpha + a)x + by + cz = 0$, $ax + (\beta + b)y + cz = 0$, $ax + by + (\delta + c)z = 0$ has a non trivial solution, then the value of $\frac{a}{\alpha}+\frac{b}{\beta}+\frac{c}{\delta}$ is
(A) –1
(B) 0
(C) 3
(D) 2

69. If A is a non-singular matrix such that $AA^T = A^TA$ and $B = A^{-1}A^T$, then matrix B is
(A) involuntary
(B) orthogonal
(C) idempotent
(D) none of these

70. Three randomly chosen non-negative integers a, b, c are found to satisfy the equation $a + b + c = 10$, then the probability that c is not even, is
(A) $\frac{1}{2}$
(B) $\frac{36}{55}$
(C) $\frac{6}{11}$
(D) $\frac{5}{11}$

71. The value of the definite integral $\int_{-1}^{1}(1+x)^{\frac{1}{2}}(1-x)^{\frac{3}{2}}dx$ equals
(A) π
(B) $\frac{3\pi}{4}$
(C) $\frac{\pi}{4}$
(D) $\frac{\pi}{2}$

72. $\int \frac{x^2-1}{x^3\sqrt{2x^4-2x^2+1}}dx$ is equal to

(A) $\frac{\sqrt{2x^4-2x^2+1}}{x^3}+c$

(B) $\frac{\sqrt{2x^4-2x^2+1}}{x}+c$

(C) $\frac{\sqrt{2x^4-2x^2+1}}{x^2}+c$

(D) $\frac{\sqrt{2x^4-2x^2+1}}{2x^2}+c$

73. The area (in sq. units) of the region $\{(x, y) : y^2 \geq 2x \text{ and } x^2 + y^2 \leq 4x, x \geq 0, y \geq 0\}$ is

(A) $\pi-\frac{8}{3}$

(B) $\pi-\frac{4\sqrt{2}}{3}$

(C) $\frac{\pi}{2}-\frac{2\sqrt{2}}{3}$

(D) $\pi-\frac{4}{3}$

74. The solution of differential equation $(2y + xy^3)dx + (x + x^2y^2)dy = 0$ is

(A) $x^2y+\frac{x^3y^3}{3}=c$

(B) $xy^2+\frac{x^3y^3}{3}=c$

(C) $x^2y+\frac{x^4y^4}{4}=c$

(D) none of these

75. A continuous and differentiable function $y = f(x)$ is such that it's graph cuts line $y = mx + c$ at n distinct points. Then the minimum number of points at which $f''(x) = 0$ is

(A) n – 1

(B) n – 3

(C) n – 2

(D) cannot say

76. The value of $\lim_{n\to\infty} \cos\left(\pi\sqrt{n^2+n}\right)$ where n is an integer

(A) $\frac{\pi}{6}$

(B) $\frac{\pi}{3}$

(C) $\frac{\pi}{4}$

(D) 0

77. If the pairs of lines $x^2 + 2xy + ay^2 = 0$ and $ax^2 + 2xy + y^2 = 0$ have exactly one line in common, then the joint equation of the other two lines is given by
(A) $3x^2 + 8xy - 3y^2 = 0$
(B) $3x^2 + 10xy + 3y^2 = 0$
(C) $y^2 + 2xy - 3x^2 = 0$
(D) $x^2 + 2xy - 3y^2 = 0$

78. Normals at two points (x_1, y_1) and (x_2, y_2) of parabola $y^2 = 4x$ meet again on the parabola where $x_1 + x_2 = 4$, then $|y_1 + y_2|$ is equal to
(A) $\sqrt{2}$
(B) $2\sqrt{2}$
(C) $4\sqrt{2}$
(D) none of these

79. An ellipse is sliding along the coordinate axes. If the foci of the ellipse are (1, 1) and (3, 3), then area of the director circle of the ellipse (in sq. units) is
(A) 2π
(B) 4π
(C) 6π
(D) 8π

80. The family of the curves which intersects the family of rectangular hyperbola $xy = c^2$ orthogonally is
(A) family of parabola
(B) family of ellipse
(C) family of circle
(D) family of rectangular hyperbola

SECTION – B
(Numerical Answer Type)

This section contains **10** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

81. If the circles $x^2 + y^2 + (3 + \sin\beta)x + (2\cos\alpha)y = 0$ and $x^2 + y^2 + (2\cos\alpha)x + 2cy = 0$ touches each other, then the maximum value of c is equal to

82. If $f(\theta) = \dfrac{1 - \sin 2\theta + \cos 2\theta}{2\cos 2\theta}$, then the value of $f(11°)\cdot f(34°)$ is equal to

83. If $0 \le x < 2\pi$, then the number of real values of x, which satisfy the equation $\sin\left(\dfrac{\pi - 2x}{2}\right) + \sin\left(\dfrac{\pi - 4x}{2}\right) + \sin\left(\dfrac{\pi - 6x}{2}\right) + \sin\left(\dfrac{\pi - 8x}{2}\right) = 0$ is equal to

84. For $n \in N$, if $\tan^{-1}\left(\dfrac{1}{3}\right) + \tan^{-1}\left(\dfrac{1}{4}\right) + \tan^{-1}\left(\dfrac{1}{5}\right) + \tan^{-1}\left(\dfrac{1}{n}\right) = \dfrac{\pi}{4}$, then n is equal to

85. If the equation $|x - 2| - |x + 1| = k$ has exactly one solution, then number of integral values of k is equal to

86. A variable plane is at a constant distance p from the origin and meets the axes in A, B and C. If the locus of centroid of the tetrahedron OABC is $\frac{1}{x^2}+\frac{1}{y^2}+\frac{1}{z^2}=\frac{\lambda}{p^2}$, then the value of λ^2 is equal to

87. The edges of a parallelepiped are of unit length and are parallel to non-coplanar unit vectors $\vec{a}, \vec{b}$ and $\vec{c}$ such that $\vec{a}\cdot\vec{b}=\vec{a}\cdot\vec{c}=\vec{b}\cdot\vec{c}=\frac{1}{2}$. Then the volume of the parallelepiped is equal to

88. If $\lim\limits_{n\to\infty}\left[(1+n)^{\frac{1}{n}}\left(1+\frac{n}{2}\right)^{\frac{2}{n}}\left(1+\frac{n}{3}\right)^{\frac{3}{n}}.....2\right]^{\frac{1}{n}}$ is L, then $\log_e L$ is equal to

89. Let $\alpha, \beta, \delta \in R$. If $f(x) = \alpha x^2 + \beta x + \delta$ is such that $\alpha + \beta + \delta = 3$ and $f(x + y) = f(x) + f(y) + xy \ \forall\ x, y \in R$, then $\sum\limits_{n=1}^{12} f(n)$ is equal to

90. If the normal to the parabola $y^2 = 4ax$ at point $P(t_1)$ cuts the parabola again at point $P(t_2)$, then least integral value of t_2^2 is equal to